

Machine Learning–based Late Delivery Risk Prediction in Global Supply Chain Operations

Research Paper | Unified Mentor Project

Abstract

This project develops a machine-learning system for predicting whether an order is likely to be delivered late before shipment. The workflow covers data preprocessing, feature engineering, supervised classification, probability-based risk classification, explainability, and deployment through a Streamlit dashboard.

1. Introduction

Late deliveries can affect customer satisfaction and operational planning. This project addresses the opportunity to identify elevated delay risk before shipment so that operations teams can review potentially problematic orders and consider proactive action.

2. Problem Statement and Objectives

Problem: Predict whether an order will be delivered late before it is shipped.

Objectives: preprocess the logistics data; engineer operational risk features; compare Logistic Regression, Random Forest, and XGBoost; evaluate predictive performance; classify probability into operational risk categories; identify predictive drivers; and deploy an interactive dashboard.

3. Dataset

The dataset contains 180,519 records and 40 original columns covering shipping, customer, product, sales, market, and regional information. The target variable is *Late_delivery_risk*.

Target	Records	Share
0 – Not late	81,542	45.17%
1 – Late risk	98,977	54.83%

The target is mildly imbalanced. Class weights were used rather than SMOTE in the final modelling workflow.

4. Data Preprocessing

Missing values occurred in Customer Lname (8) and Customer Zipcode (3); these were filled using the respective modes. No duplicate rows were identified. Whitespace was removed from categorical fields. Post-shipment/text fields were excluded where appropriate. Categorical variables were one-hot encoded and numerical variables were scaled within the training pipeline. An 80/20 stratified train-test split was used.

5. Feature Engineering

Shipping Pressure Index: $\text{Order Item Quantity} / (\text{Days for shipment (scheduled)} + 1)$.

Mode flags: First Class, Second Class, Same Day, Standard Class.

Regional Congestion Indicator: order count for the Order Region.

Order Complexity Score: normalized combination of quantity, discount rate, and shipping pressure.

6. Model Development and Evaluation

Three classifiers were evaluated.

Model	ROC-AUC	Precision	Recall	F1
Logistic Regression	0.7430	0.8463	0.5439	0.6622

Model	ROC-AUC	Precision	Recall	F1
Random Forest (initial)	0.8171	0.8483	0.6158	0.7136
XGBoost	0.7843	0.8362	0.5755	0.6817
Random Forest (cleaned/deployed)	0.7598	0.7947	0.6075	0.6886

The initial Random Forest showed high importance for identifiers and location-proxy variables. Customer Id, Order Customer Id, Category Id, Department Id, Customer Zipcode, Latitude, and Longitude were removed. The cleaned model therefore trades some predictive performance for a more defensible operational feature set.

7. Risk Probability and Classification

The deployed model produces a probability from 0 to 1. Initial operational categories are Low Risk below 0.40, Medium Risk from 0.40 through 0.70, and High Risk above 0.70. These are operational starting thresholds, not statistically optimized cutoffs.

8. Explainability

Leading operational feature-importance signals included Benefit per order, Order Profit Per Order, Order Item Profit Ratio, Days for shipment (scheduled), Order Item Total, Sales per customer, Order Item Discount, Standard Class indicators, Order Complexity Score, Discount Rate, Regional Congestion Indicator, and Shipping Pressure Index. These are predictive signals rather than causal explanations.

9. Deployment

The Streamlit dashboard provides Delay Risk Overview, Region & Mode Risk Analysis, Order-Level Risk Prediction, and an Operations Action Panel. It includes filters for Shipping Mode, Market, Customer Segment, and risk threshold. The deployed Random Forest was reduced from 200 trees to 50 trees for cloud resource efficiency, with a compressed model size of approximately 52.43 MB.

Live dashboard: <https://late-delivery.streamlit.app/>

10. Insights and Recommendations

Use high-risk predictions to prioritize proactive operational review; monitor medium-risk orders and available capacity; and continue standard processing for low-risk orders. Region and shipping-mode views can support investigation of historical risk patterns. Model output should be combined with current logistics conditions and human judgment.

11. Limitations and Future Improvements

The model is predictive rather than causal. Risk thresholds were not optimized against explicit intervention costs. Derived aggregate features should ideally be computed strictly from training-period information in a production setting. Future improvements include probability calibration, cost-sensitive threshold optimization, time-based validation, model-drift monitoring, richer explainability, and additional operational signals.

12. Conclusion

The project demonstrates an end-to-end workflow from logistics data preparation through machine-learning prediction, explainability, and cloud deployment. The resulting dashboard provides an accessible prototype for pre-shipment late-delivery risk assessment.